

T4A2: Cutoff of the Lennard-Jones potential

The Lennard-Jones potential is

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right],$$

with $\epsilon > 0$, $\sigma > 0$ and $r > 0$.

a) Cutoff where all attractions are neglected

Attraction is about the force, not about the potential. So I first need the force

$$F(r) = -\frac{dV}{dr}.$$

F is the force along the line between the two particles. $F > 0$ means they are pushed apart (repulsion) and $F < 0$ means they are pulled together (attraction).

I write $V(r) = 4\epsilon (\sigma^{12}r^{-12} - \sigma^6r^{-6})$ and use $\frac{d}{dr}r^n = nr^{n-1}$:

$$\frac{dV}{dr} = 4\epsilon \left(-\frac{12\sigma^{12}}{r^{13}} + \frac{6\sigma^6}{r^7} \right).$$

So the force is

$$F(r) = 4\epsilon \left(\frac{12\sigma^{12}}{r^{13}} - \frac{6\sigma^6}{r^7} \right) = \frac{24\epsilon}{r} \left(\frac{\sigma}{r} \right)^6 \left[2 \left(\frac{\sigma}{r} \right)^6 - 1 \right].$$

Since $24\epsilon/r > 0$ and $(\sigma/r)^6 > 0$, the sign of F is the sign of $2(\sigma/r)^6 - 1$. $F = 0$ when $(\sigma/r)^6 = 1/2$, so $r^6 = 2\sigma^6$ and $r = 2^{1/6}\sigma$. For $r < 2^{1/6}\sigma$ we have $(\sigma/r)^6 > 1/2$, so $F > 0$, which is repulsion. For $r > 2^{1/6}\sigma$ we have $(\sigma/r)^6 < 1/2$, so $F < 0$, which is attraction. Therefore, all the attraction is at $r > 2^{1/6}\sigma$. With the cutoff

$$r_c = 2^{1/6}\sigma \approx 1.12\sigma$$

all the attractions are neglected and only the repulsion is left.

Note that it is not $r_c = \sigma$. At $r = \sigma$ the potential is zero, but $F(\sigma) = \frac{24\epsilon}{\sigma}(2 - 1) = \frac{24\epsilon}{\sigma} > 0$, so the force is still repulsive. Between σ and $2^{1/6}\sigma$ the potential is negative, but the particles are still pushed apart.

b) Truncated and shifted potential

A cutoff sets $V = 0$ for $r \geq r_c$. If I only do that, the potential jumps at r_c from $V(r_c)$ to 0. $V(r_c)$ is not zero: for $r > \sigma$ we have $(\sigma/r)^{12} < (\sigma/r)^6$, so $V(r) < 0$. A jump in V means that a particle crossing r_c suddenly gains or loses energy, so the total energy is not conserved.

To fix this, I shift the potential inside the cutoff by $V(r_c)$:

$$V_{\text{cut}}(r) = \begin{cases} V(r) - V(r_c), & r < r_c, \\ 0, & r \geq r_c. \end{cases}$$

For the LJ potential this is, for $r < r_c$,

$$V_{\text{cut}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - 4\varepsilon \left[\left(\frac{\sigma}{r_c} \right)^{12} - \left(\frac{\sigma}{r_c} \right)^6 \right].$$

V is continuous for $r > 0$, so coming from $r < r_c$:

$$\lim_{r \rightarrow r_c^-} V_{\text{cut}}(r) = V(r_c) - V(r_c) = 0,$$

which is the same as $V_{\text{cut}}(r) = 0$ for $r \geq r_c$. Therefore, V_{cut} is continuous at r_c .

The shift does not change the force. For $r < r_c$, $V(r_c)$ is a constant, so its derivative is zero:

$$F_{\text{cut}}(r) = -\frac{dV_{\text{cut}}}{dr} = -\frac{dV}{dr} + \frac{d}{dr}V(r_c) = F(r), \quad r < r_c,$$

and $F_{\text{cut}} = 0$ for $r \geq r_c$.

With the cutoff from a), $r_c = 2^{1/6}\sigma$, we have $(\sigma/r_c)^6 = 1/2$ and $(\sigma/r_c)^{12} = 1/4$, so $V(r_c) = 4\varepsilon(1/4 - 1/2) = -\varepsilon$. The modified potential is then $V(r) + \varepsilon$ for $r < 2^{1/6}\sigma$ and 0 outside, which is purely repulsive.

c) Potential and force continuous at any cutoff

In b) the force for $r < r_c$ is still $F(r)$, and for $r \geq r_c$ it is 0. So the force jumps at r_c from $F(r_c)$ to 0. From a), $F(r_c) = 0$ only for $r_c = 2^{1/6}\sigma$, so for every other cutoff the force is discontinuous. A jump in the force means the force changes suddenly between two time steps of the Verlet algorithm, so the total energy is not conserved well.

To fix this, I also subtract the tangent line of V at r_c , so that both the value and the slope are zero at r_c . This is the shifted-force potential:

$$V_{\text{sf}}(r) = \begin{cases} V(r) - V(r_c) - (r - r_c) V'(r_c), & r < r_c, \\ 0, & r \geq r_c, \end{cases}$$

where $V'(r_c) = \frac{dV}{dr}|_{r=r_c}$ is a constant number, because r_c is fixed. Using $V'(r_c) = -F(r_c)$ and F from a), for the LJ potential this is, for $r < r_c$,

$$\begin{aligned} V_{\text{sf}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - 4\varepsilon \left[\left(\frac{\sigma}{r_c} \right)^{12} - \left(\frac{\sigma}{r_c} \right)^6 \right] \\ + (r - r_c) \frac{24\varepsilon}{r_c} \left[2 \left(\frac{\sigma}{r_c} \right)^{12} - \left(\frac{\sigma}{r_c} \right)^6 \right]. \end{aligned}$$

The potential is continuous at r_c :

$$\lim_{r \rightarrow r_c^-} V_{\text{sf}}(r) = V(r_c) - V(r_c) - (r_c - r_c) V'(r_c) = 0,$$

which is the same as for $r \geq r_c$.

The force is also continuous at r_c . For $r < r_c$, $V(r_c)$ is a constant, so its derivative is zero, and $(r - r_c)V'(r_c)$ is a straight line with slope $V'(r_c)$. So

$$\frac{dV_{\text{sf}}}{dr} = V'(r) - V'(r_c),$$

and the force is

$$F_{\text{sf}}(r) = -\frac{dV_{\text{sf}}}{dr} = F(r) - F(r_c), \quad r < r_c.$$

For the LJ potential this is

$$F_{\text{sf}}(r) = \frac{24\varepsilon}{r} \left[2 \left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - \frac{24\varepsilon}{r_c} \left[2 \left(\frac{\sigma}{r_c} \right)^{12} - \left(\frac{\sigma}{r_c} \right)^6 \right], \quad r < r_c.$$

Then

$$\lim_{r \rightarrow r_c^-} F_{\text{sf}}(r) = F(r_c) - F(r_c) = 0,$$

which is the same as the force for $r \geq r_c$. Therefore, both V_{sf} and F_{sf} are continuous at r_c . This works for any r_c , because I subtract exactly the value $V(r_c)$ and the slope $V'(r_c)$, whatever they are.

The price is that the force inside the cutoff is changed by the constant $F(r_c)$. For large r_c , $(\sigma/r_c)^{12}$ is much smaller than $(\sigma/r_c)^6$, so

$$F(r_c) \approx -\frac{24\varepsilon\sigma^6}{r_c^7},$$

which goes to zero quickly when r_c grows. So for a large enough r_c this is a small perturbation. For $r_c = 2^{1/6}\sigma$ from a), $F(r_c) = 0$, and V_{sf} is the same as V_{cut} from b).